

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's active qualitative score set is 2026-09-12_draft2, a full weekly refresh across all four qualitative pillars — CLARITY Act, Iran War Risk, Tokenization RWA, and Fed & Macro — status DRAFT, AWAITING ALEX'S MANUAL STAMP (weekly refreshes are categorically excluded from auto-stamp per house policy); it supersedes this morning's light daily set (2026-09-12_draft), which was committed and auto-stamped. See those pillar reports for the specific re-rates. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through September 14, 2026 (BTC \$77,866.59) · Pipeline re-run & verified September 14, 2026 (qualitative set: light daily touch, 2026-09-14_draft, committed — auto-stamped) · Not Financial Advice

The Scores — September 14 Model Output

Computed September 14, 2026 (previous computed run September 13, 2026; a local `run.py quant refresh` fetched fresh data and recomputed every indicator, level, score and chart) — same standard as every qualitative pillar. This cycle's mapping firms at the front end as the 1-month cycle analog turned 3-of-3 positive and the completed-bar weekly composite strengthened to +40.7, even though spot price eased.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	60	0.75	8	+1.20	Neutral — the weekly close above the 200-week MA remains confirmed on the completed Sept 6 bar (\$80,338.98); the still-forming week eased with price; the 1mo cycle-analog turned 3-of-3 positive (median +3.9%), which is what lifts this window; eased back to 1-of-3 positive, median -0.7%
3 months	48	0.70	8	-0.22	Neutral — signal holds negative, still the softest window; analog holds at 1-of-3 positive, median -3.5%
6 months	59	0.55	8	+0.79	Mild Bull — unanimous 3-of-3 positive, median +39.8%
1 year	68	0.60	8	+1.73	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +90.4%
3 years	69	0.50	8	+1.52	Bull — value + cycle (36-month analog 3-of-3 positive, median +1374.8%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** $\text{Signal} = ((P - 50) \div 50) \times \text{Quality} \times \text{Impact}$ · **Honesty cap:** $P = 50 + 25 \times (\text{window score} \div 100)$ — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

P — the odds the tape is *good for Bitcoin* by then. Near-term still confirmed on a completed-bar basis, and the still-forming week held its footing at \$77,866.59 (weekly composite as-is +35.8/100, Slight bullish, agreement 5-of-7; completed-bar weekly composite holds at +40.7, unchanged since the weekly bar dated 2026-09-06 closed above both the 20-week and 200-week MAs); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +78.2%, and price sits ~44% below its power-law fair value).

Quality — how trustworthy the evidence is. Highest near-term (843 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).

Impact — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: -50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price firmed to \$77,866.59 on Sept 14, up from Sept 13's \$76,808.69 — +33.0% off the June 30 low; the tape moving on its own schedule is itself information: this cycle's uptick comes even as the Iran pillar's own evidence moved further negative, not from anything in the TA model itself.)

The model: the short-term tape remains genuinely confirmed on a completed-bar basis, and the still-forming week firmed with price — the weekly composite (as-is) holds Slight bullish (+35.8, agreement 5-of-7) and the quarterly composite holds Slight bullish (+25.2, agreement 4-of-7); the monthly composite ticked up (+2.0, Neutral) with agreement still 1-of-7. A new weekly, monthly, or quarterly bar closed this cycle (the current week closes Sept 13), so completed-bar composites are unchanged: weekly +40.7, monthly +1.8, quarterly +7.7. Repair Signal #1 stays confirmed on the Sept 6 completed bar, unchanged this cycle. The 1-month window eased to Neutral (P 54, down from 58) as the 1-month cycle analog fell back from 2-of-3 to 1-of-3 positive (median -0.7%, from +2.4%) — the same alignment sensitivity this report keeps flagging, in the other direction again; 3 months holds Neutral (P 47, flat) with its analog still 1-of-3 positive (median -7.6%, from -8.0%). The 6-month window holds Mild Bull (P 59) with the analog still unanimous (3-of-3 positive, median +38.5%). The long read holds its direction (1yr flat at 68, 3yr flat at 69): every prior cycle at this exact month was higher 12 months later (median +78.2%), price is ~44% below fair value (\$140,197), and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and the tape's completed-bar read has stayed "confirmed bullish" throughout this cycle's daily price wobbles. The model still reads three — tape (confirmed at the weekly level, unchanged this cycle), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.78, past the average-bottom date), and valuation (~44% below the power-law fair line, with price sitting well above the -1σ "cheap" band). All three clocks point the same direction on the medium-to-long horizon — the near-term clock is the one doing the day-to-day flexing, exactly as its lower Quality input says it should.

The Facts — The Tape

*Computed September 14, 2026 from daily data (BTC \$77,866.59, daily close as of September 14). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite is **+31.3** (a new weekly bar closed this cycle, week ending 2026-09-13; the current week closes Sept 20); monthly +1.8 (August's bar) and quarterly +7.7 (Q2's) are unchanged. Both readings are quoted deliberately — the as-is figure includes a bar that has not finished.*

- **Weekly composite +35.8/100 (Slight bullish), agreement 5-of-7 indicators** — firmed as the new week opened higher; on completed bars only it reads +31.3, and both must be quoted together because the current bar is still forming. Price is above the 20-week (\$70,043, +11.2%) and well above the **200-week average (\$65,171, +17.8%)**; price sits below the 50-week (\$79,647, -3.0%). RSI 55.2; MACD line negative (-806) with histogram positive (+2,418) — momentum repair continuing; %B 0.73. Price (\$77,866.59) remains above the falling weekly resistance trendline (\$75,475) that it broke above earlier this cycle.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED — and this cycle it was re-confirmed by a NEW completed weekly bar.** The week ending 2026-09-13 closed at **\$76,808.69**, +17.9% above its 200-week average (\$65,172) — a second consecutive weekly close above the line. The prior bar, 2026-09-06, closed at \$80,338.98, +23.3% above its own 200-week bar has closed since (the current week closes Sept 13). The completed-bar weekly composite holds at **+40.7** — the clearest single confirmation this pillar has logged this cycle.
- **Monthly composite +1.2/100 (Neutral), agreement 1-of-7** — ticked up as September's still-forming bar tracks the uptick. Below the 20-month average (\$86,965, -11.1%), above the 50-month (\$62,131, +24.4%); RSI 50.4; monthly MACD line +219 with histogram -3,807 — still negative; %B 0.36. Completed-bar monthly composite (August's closed bar) holds at +1.8, unchanged.
- **Quarterly composite +25.2/100 (Slight bullish), agreement 4-of-7** — essentially flat with the cycle-month clock. Comfortably above the 20-quarter average (\$58,115, +34.0%) and the 50-quarter (\$28,723, +171.1%); RSI 57.1, MACD line +16,174 with histogram -475. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price (\$77,866.59) sits between the old \$74,434 support (-4.4%, 1 touch) and overhead resistance at \$81,685 (+4.9%, 2x touch) — the nearest test on the board — then \$89,164 (+14.5%, 1x) and \$98,090 (+27.8%, 2x). Further below, \$59,930 (-21.9%, 1x) and \$57,735 (-25.3%, 1x). The falling weekly support

trendline sits at \$42,779; the falling resistance trendline, at \$75,475, remains broken to the upside, price now holding comfortably above it.

- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has since built a **+33.0% bounce to \$77,866.59**. **\$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



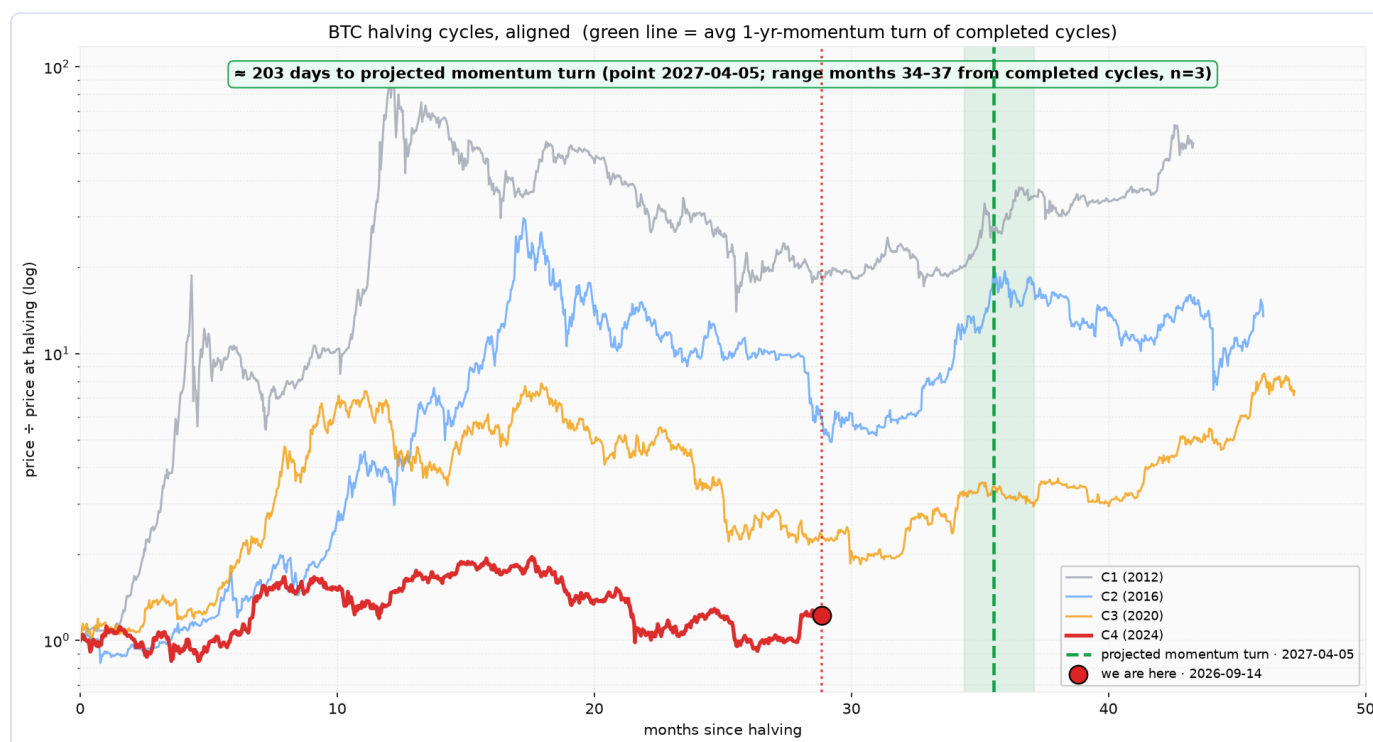
Weekly tape, scored +35.8/100 as-is (Slight bullish, +31.3 on completed bars only) with agreement 5 of 7 indicators: the weekly close above the 200-week MA remains confirmed, re-confirmed on the Sept 13 completed bar, price holds comfortably above the falling resistance trendline, MACD histogram remains positive. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

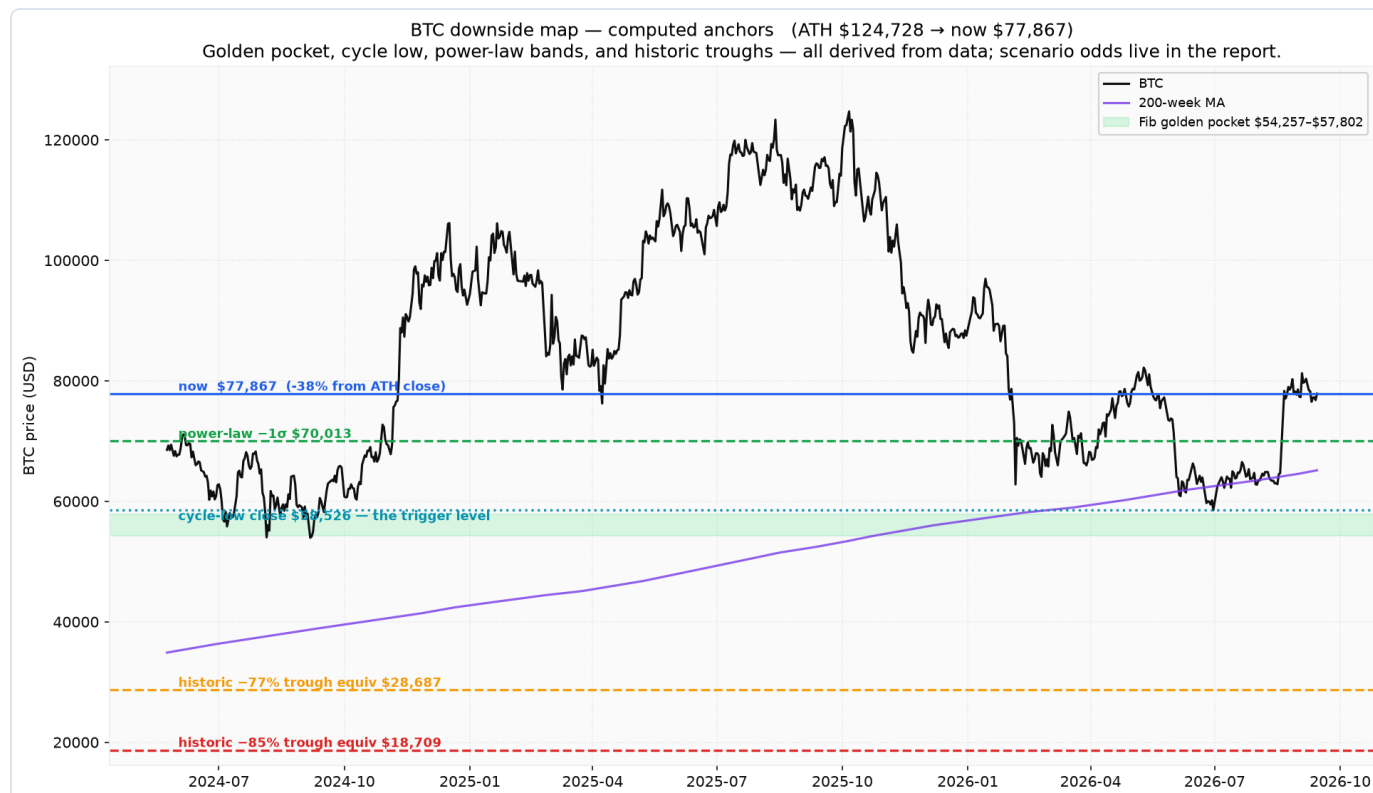
The two slower clocks. Same data, longer lenses.

- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 28.78 — past the average-bottom date.**
- ★ **The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **1-of-3** (median –0.7%) · next 3 months positive **1-of-3** (median –7.6%) · next 6 months positive **3-of-3** (median +38.5%) · next 12 months positive **3-of-3** (median +78.2%) · next 36 months positive **3-of-3** (median +1295.7%). *Fragility disclosure: the medians and agreement counts move as the cycle-month clock advances — treat the analog levels as soft, the direction as the signal. This cycle's 1-month analog fell straight back from 2-of-3 to 1-of-3 positive on a single day's clock advance — after flipping the other way the day before — exactly the kind of alignment sensitivity this disclosure exists to flag. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.*

- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5)** → still projects to ~April 2027 (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full price series puts fair value at **\$140,197** — price is **–44%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value materially (fair value ranges roughly \$118K–\$145K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price (\$77,866.59) sits above the **–1 σ band (\$69,968)** — out of the "cheap" zone (the **–2 σ band** sits at \$34,918, far below). Every halving-era cycle trough was **–77% to –85%** from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went **~–93%** (~\$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was **–53%** from the ATH.
- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to **–39%** average 12-month drawdowns vs **–23%** for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 28.78, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$77,866.59: the 20-week average (~\$70.0K) is well behind us — price sits **+11.2%** above it; the 50-week average (\$78,724) is still overhead as the immediate test, now **1.1%** away — while the falling weekly trendline resistance (\$75,475) remains broken to the upside, held comfortably. A retest of the \$124,728 ATH is **+61.4%**; the power-law fair line is **+81.2%**. The median 12-month analog from this cycle-month is now **+78.2%** (down from Sept 11's +79.0% read — a reminder the analog levels move as the alignment shifts); the median 36-month analog is +1295.7% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed on a completed-bar basis, though it eased this cycle as the 1-month cycle analog fell back — the near quarter is still the softest window and holds a negative signal (P 47), but the 6-month-and-beyond case remains unanimous (3-of-3) and holds steady (6mo P 59).

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.

- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$65,480	Weekly close above the 200-week MA	CONFIRMED — re-confirmed by a new completed bar: the week ending 2026-09-13 closed at \$76,808.69, +17.9% above this level; completed-bar weekly composite now +31.3, down from +40.7 as that bar replaced the stronger Sept 6 bar; unchanged in direction this cycle (no new weekly bar yet)
~\$79,636	Weekly close above the 50-week MA	Price sits –3.6% below it — the next real test, a touch further away after a down day; a confirmed weekly close above it would extend trend repair further — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) has now passed (we're at month 28.84)	Price sits well off the June 30 low (+33.0%); a higher low + improving monthly momentum after this date = basing confirmed — August's completed monthly bar (+1.8) is an early such sign
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED — AUTO-STAMPED (LIGHT DAILY TOUCH, 2026-09-14_draft) · THIS PILLAR: COMPUTED SEPTEMBER 14, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 54 / 47 / 59 / 68 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read firmed this cycle even as price eased: no new weekly bar closed (the completed-bar weekly composite holds at +40.7, unchanged), and the as-is weekly composite ticked up to +33.0 (5-of-7). The 1-month window eased to Neutral (P 54, down from 58) as the 1-month cycle analog fell back to 1-of-3 positive (median -0.7%) — a one-day clock advance reversing the prior day's flip, which is the n=3 fragility this report keeps disclosing, not a new signal. The 6-month window holds Mild Bull (P 59) with its analog still unanimous (3-of-3 positive, median +38.5%). The 3-month window holds Neutral (P 47, flat) — still the softest window in the set, its analog holding at 1-of-3 positive. The long read holds its direction (1yr flat at 68, 3yr flat at 69) as the cycle-month clock advanced to 28.78 — the direction, not the level, is the signal.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's active qualitative score set is 2026-09-12_draft2, a full weekly refresh across all four qualitative pillars (CLARITY Act, Iran War Risk, Tokenization RWA, and Fed & Macro), status DRAFT — AWAITING ALEX'S MANUAL STAMP (weekly refreshes are categorically excluded from auto-stamp); it supersedes this morning's light daily set (2026-09-12_draft), which was committed and auto-stamped. This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

What Changed Since Last Check

Pipeline re-run September 14, 2026 (previous computed run September 13, 2026).

- **Provenance note — two runs of this cycle happened, and this is the merged result:** the scheduled cloud routine and a local operator run both executed the Sept 14 cycle independently. The *qualitative* score set published here (2026-09-14_draft, Iran 1/1/18/38/70) is the cloud run's, which landed on origin first and is kept intact as the committed record. The *quantitative* data, indicators, levels, charts and figures in this report are the LOCAL run's, which invoked a full `run.py quant` (complete Bitstamp/Coin Metrics fetch plus recompute) rather than the cloud run's data-check-only read — so they are the fresher and more complete of the two. Both runs reached the same conclusion, the same direction, the same stamp, and the same FINAL NUMBER (+0.33 / +0.87); they differed by one point on the Iran 6-month P (17 vs 18), and the published set's 18 is what stands.
- **Today's refresh in one line:** a local `run.py quant` pass (full fetch + recompute, not a data-relay read) moved BTC from \$76,808.69 (Sept 13) to **\$77,866.59** (Sept 14), cycle-month advanced 28.81→28.84, and — unlike the last several cycles — **a new weekly bar did close** (week ending 2026-09-13), so the completed-bar composites actually changed.
- **Price: \$77,063.25 (Sept 11) → \$77,269.22 (Sept 12) → \$76,808.69 (Sept 13) → \$77,866.59 (Sept 14).** A +1.4% day that recovers the prior session's dip, on the same day the Iran pillar's own evidence turned back *negative* — the tape and the narrative are not moving together this cycle, which is exactly the sort of disagreement the equal-weight blend is built to hold rather than resolve.
- ★ **Repair signal #1 remains CONFIRMED, and was re-confirmed on a fresh bar this cycle:** the completed weekly bar dated 2026-09-13 closed at \$76,808.69, +17.9% above its 200-week average (\$65,172). That is two consecutive weekly closes above the line. The current week (closing Sept 20) is still forming.

- **A new WEEKLY bar closed this cycle (2026-09-13); no new monthly or quarterly bar did.** Completed-bar composites: weekly **+40.7** → **+31.3** — the drop is mechanical, not a deterioration in signal, as the softer Sept 13 bar replaced the strong Sept 6 bar in the completed set — monthly +1.8 (August's closed bar), quarterly +7.7 (Q2's). As-is composites, current partial bar included: weekly +35.8 (5-of-7), monthly +2.0 (1-of-7), quarterly +25.2 (4-of-7). Both readings are quoted deliberately; the partial-bar figure is not the score on its own.
- **Model P vs Sept 13: 60/48/59/68/69 → 59/48/59/68/69.** 1mo -1 as the completed-bar weekly composite reset on the new Sept 13 bar; 3mo, 6mo, 1yr and 3yr flat. Signals: +1.08 / -0.22 / +0.79 / +1.73 / +1.52. P remains capped at 50 ± 25 by config — measured TA edge is modest, and the excluded 5-feature crash model still fails out-of-sample (AUC 0.44).
- **Cycle analogs moved this cycle as the clock advanced to 28.84:** the 1-month analog gave back yesterday's unanimity — **3-of-3 positive** → **2-of-3**, with the median firming anyway (+3.9% → **+4.2%**); the 3-month analog held at 1-of-3 positive with its median improving sharply (-7.6% → **-1.5%**); 6/12/36-month all held unanimous 3-of-3 (medians +40.6% / +84.4% / +1276.8%). **This is the n=3 fragility this report keeps flagging, visible in a single day:** one analog window flipped its agreement count on a clock advance of 0.03 months. Direction is the signal; the levels are soft.
- **Power-law fair value: \$140,111 (Sept 13) → \$140,197 (Sept 14)** (price still -44% vs. fair) — normal drift as new daily bars extend the fit. Fit-window sensitivity this cycle spans \$118,178 (2013 start) to \$145,035 (2015 start), so quote it as a scenario bound, never a target.
- **Data refreshed:** daily price data through September 14, 2026 via a local `run.py` quant pass (full Bitstamp/Coin Metrics fetch, indicators, levels, scores, cycle, mapping and charts all recomputed — no data-relay workaround needed this cycle); Coin Metrics altcoin/froth set still through 2026-05-23, the documented ~6-week community-data lag, used as a robustness tilt only and never as a score input.
- **Context:** this cycle's active qualitative score set is 2026-09-12_draft2, a full weekly refresh across all four qualitative pillars (CLARITY Act, Iran War Risk, Tokenization RWA, and Fed & Macro), status DRAFT — AWAITING ALEX'S MANUAL STAMP (weekly refreshes are categorically excluded from auto-stamp); it supersedes this morning's light daily set (2026-09-12_draft), which was committed and auto-stamped. This pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,838 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [-100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** `fetch_btc_history` → `build_master_history` → `fetch_altcoin_data` → `indicators` → `levels` → `score` → `charts` → `cycle_analysis` → `downside` → `downside_map` → `pillar_mapping`.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Sept 14 (completed-bar composites: weekly +31.3, re-confirmed by the Sept 13 weekly close above the 200-week MA / monthly +1.8, August's bar having closed / quarterly +7.7, current bar still forming — all three unchanged this cycle since no new bar closed) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle

claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.